

EX. No: 2a)

Shell Script

AIM:

To write a Shellscript to display basic calculator.

```
1 #!/bin/bash
2
3 echo "Enter first number"
4 read a
5 echo "Enter second number"
6 read b
7 echo "1.ADD 2.SUBTRACT 3.MULTIPLY 4.DIVISION 5.MODULO"
8 read ch
9 case $ch in
10     1) echo "Result = $(expr $a + $b)" ;;
11     2) echo "Result = $(expr $a - $b)" ;;
12     3) echo "Result = $(expr $a * $b)" ;;
13     4) echo "Result = $(expr $a / $b)" ;;
14     5) echo "Result = $(expr $a % $b)" ;;
15     *) echo "Invalid choice" ;;
16 esac
```

```
[PraveenKumar212@localhost ~]$ mkdir shell_script
[PraveenKumar212@localhost ~]$ cd shell_script
[PraveenKumar212@localhost shell_script]$ touch calculator.sh
[PraveenKumar212@localhost shell_script]$ vi calculator.sh
[PraveenKumar212@localhost shell_script]$ bash calculator.sh
Enter first number
[PraveenKumar212@localhost shell_script]$ vi calculator.sh
[PraveenKumar212@localhost shell_script]$ bash calculator.sh
Enter first number
5
Enter second number
5
1.ADD 2.SUBTRACT 3.MULTIPLY 4.DIVISION 5.MODULO
+
Invalid choice
[PraveenKumar212@localhost shell_script]$ bash calculator.sh
Enter first number
6
Enter second number
7
1.ADD 2.SUBTRACT 3.MULTIPLY 4.DIVISION 5.MODULO
1
Result = 13
[PraveenKumar212@localhost shell_script]$
```

Ex. No: 2b)

Shell Script

AIM:

To write a Shellscrip to test given year is leap or not using conditional statement.

```
1 echo "Enter a year"
2 read y
3 if [ $((y%400)) -eq 0 ] || [ $((y%4)) -eq 0 -a $((y%100)) -ne 0 ]
4 then
5     echo "Leap year"
6 else
7     echo "Not a leap year"
8 fi
```

```
[PraveenKumar212@localhost ~]$ ls
allocation Desktop Downloads Pictures shell_script Templates
awk Documents Music Public system_calls Videos
[PraveenKumar212@localhost ~]$ cd shell_script
[PraveenKumar212@localhost shell_script]$ ls
'\ calculator.sh fibonacci.sh leap_year.sh reverse.sh
[PraveenKumar212@localhost shell_script]$ bash leap_year.sh
Enter a year
2026
leap_year.sh: line 3: [: missing ']'
Not a leap year
[PraveenKumar212@localhost shell_script]$ vi leap_year.sh
[PraveenKumar212@localhost shell_script]$ bash leap_year.sh
Enter a year
2026
Not a leap year
[PraveenKumar212@localhost shell_script]$ █
```